

aegis-statefulset — Submission

Submission — aegis-statefulset

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Submission — aegis-statefulset

A reference architecture for single-master-AZ Kubernetes hosting of LevelDB-backed stateful applications, with cold DR via Velero, Strangler-Fig migration support, and observability built in.

Status: proof-of-concept. Architecture is stable; implementation is skeletal by design — the submission covers structural depth (10 thematic ADRs (50 private predecessors) across 14 domains) over surface coverage. Anonymised for portability.

0. Grounding — current public commitments vs spec target vs what this delivers

This architecture is calibrated against the customer's **publicly-stated** operational baseline + the **spec's stated improvement target**, not against a guess at the customer's internal aspiration. The numbers below are the anchor; every architectural decision in §§ 2–7 references back here.

Dimension	Current public commitment	Spec target	This architecture delivers
Uptime SLA	99.50% for Team /	(spec implicit: "automatically	99.5%+ achievable with ~25 min AZ-failure RTO target (well inside

Dimension	Current public commitment	Spec target	This architecture delivers
	Business; "SLA possible" for Enterprise ¹ _	recover from failures")	the 3.6h/month budget the SLA implies)
RPO	~24 hours — "automatic backup of databases every day" ² _	≤ 6 hours (spec, Backup & Restore section)	5-min cadence default → ~30 sec typical RPO (operator-tunable to 6h via <code>backup.cadence_minutes</code>)
RTO	Not publicly stated	Not stated in spec	~25 min AZ failure / ~50 min region (architecture's chosen target)
Backup mechanism	"regularly generates incremental backups... secured at separate locations" ³ _	"Restic backup with LVM for 2 TB of data with each pod"	Velero + EBS Snapshot + AWS Data Lifecycle Manager (DLM) cross-region copy (Velero's File System Backup (FSB) path using Restic / Kopia is documented as an opt-in patch — see <code>docs/future/restic-fsb-patch.md</code>)
DR feature	"high availability with automated fail-over and regular backups" — included across all tiers ⁴ _	"node and pod failures do not result in data loss"	Three-Layer DR (terraform / helm-via-tf / Velero); cold-DR posture; manual operator-gated rotation per <code>docs/operations/why-cold-dr.md</code>

What the grounding implies

- **The spec asks for ~4× RPO improvement** over current public commitment (24 h → 6 h). This architecture delivers ~480× via the 5-minute cadence default. This is over-delivery on the lever the spec asks about, deliberately — to demonstrate the *curve*, not just hit the *floor*. The operator picks the operating point given the customer's actual operational tolerance + cost ceiling.
- **The 99.5% uptime SLA is the architecture's binding RTO constraint.** 3.6 hours/month total downtime budget. At our chosen 25-min AZ-failure RTO, this comfortably allows 8 incidents per month. If the customer's actual incident frequency is higher (e.g., weekly AZ events), the RTO target tightens — Stage 3 conversation.
- **The current backup mechanism is incremental + cross-location** — the spec's "Restic + LVM" naming is one tooling choice for that capability. Velero + EBS Snapshot is the K8s-native equivalent that meets the same capability requirement with materially better RTO and cluster-state coverage. Reasoning chain in `docs/operations/why-velero-not-restic.md`.
- **Stage 3 question that this section directly raises:** what's the customer's *aspirational* uptime + RPO, vs the *current public* commitment? The architecture is built to support an aspirational target tighter than the current 99.5% / 24h. If the aspiration is the same as current public, much of the architecture is over-built; if the aspiration is 99.99% / 5-min RPO, the architecture is appropriately calibrated; if

99.999%, the architecture's storage primitive (LevelDB) becomes the constraint, not the platform.

1. Overview — three organising principles

Every architectural decision traces back to one of these. Every knob in `values.yaml` is the operator's seat at one of these levers.

P1 — Data on EBS is the only irreplaceable asset; everything else is declarative

Customer state lives on encrypted EBS volumes managed via LVM. Cluster, deployments, routing tables, container images, AZ topology — all reproducible in 30–60 minutes from Helm + Terraform. Stateful tier gets `reclaimPolicy=Retain`, 1:1 pod-to-node, conservative backup cadence. Stateless tier gets autoscaling, spot mix, cattle treatment.

P2 — RPO is a configurable trade-off, not a fixed floor

The challenge spec sets RPO 6h as the upper bound. The default is tighter — 5-min cadence yields typical recovery point ~30 sec — but the lever is exposed in `values.yaml` (`backup.cadence_minutes`), and the cost-RPO curve is documented for operator self-tune. The submission gives the curve, not a pre-decided answer.

P3 — Conservative on recovery, aggressive on failure detection

Failover detection: 50% unhealthy / 2 min sustained — reactive, sensitive. Recovery detection: 95% healthy / 60 min sustained — notify-only, never auto-failback. Asymmetric thresholds prevent flap.

2. Hard architectural requirements (Layer 1 — platform owns)

Layer 1 decisions are non-negotiable for the customer; the platform owns them.

#	Requirement	Why
L1.1	Single master AZ for all workloads at any moment	LevelDB is single-writer; cross-AZ chatter on a stateful tier with no replication primitives is wasted latency and double-cost. (ADR-01 modified.)
L1.2	EBS encryption at rest with per-tier customer-managed KMS keys	Compliance and blast-radius isolation. (ADR-07.)
L1.3	StatefulSet with <code>reclaimPolicy=Retain</code> +	Pod-to-PV mapping must survive cluster recreation. (ADR-02.)

#	Requirement	Why
	claimRef pre-binding	
L1.4	1:1 pod-to-node ratio for the stateful tier	Disk-bound workloads compete poorly for I/O; 1:1 is honest about it. (ADR-02.)
L1.5	Multi-PVC EBS storage (data + WAL) per pod, online-expandable via CSI volume resize	Spec named LVM; modern K8s database operators (TiDB / K8ssandra / CloudNativePG) use multi-PVC for the same WAL/data IO isolation benefit without privileged init or Kyverno PolicyException. Literal LVM available as opt-in patch (docs/future/lvm-init-patch.md). (ADR-02.)
L1.6	Cross-region S3 replication of backups	Single-region durability is not a DR posture. (ADR-04.)
L1.7	Three-path DR with named RPO/RTO per path	Operators must know which path applies for which failure shape. (ADR-04.)
L1.8	Three-Layer DR — Layer 1 (Velero) cold + Layer 2 (Terraform helm_release) warm + Layer 3 (Route 53) DNS	Each layer recovers a different failure class without contention. (ADR-04.)
L1.9	SHA-pinned external image references	Supply-chain hygiene per NIST Secure Software Development Framework (SSDF) / Supply-chain Levels for Software Artifacts (SLSA) L3. (ADR-09, ADR-09.)
L1.10	OpenTelemetry-only instrumentation	Vendor neutrality at the instrumentation layer; backend is reversible. (ADR-06, ADR-06.)

3. Demo-only choices (Layer 2 — platform decides for the POC)

These are honest defaults the platform picks for the POC submission. Each is re-decidable post-conversation; none is load-bearing on the architecture.

#	Choice	Reasoning
L2.1	Mock stateful app — ~30 lines Go, file-per-key under /data	Smallest object that exercises the operational shape (PV ownership, write-on-disk, node-loss survival). (app/main.go.)
L2.2	Pattern 2 LDB layout (one DB per pod) for the POC mock	Pattern 1 vs 2 is a Stage 3 question; Pattern 2 is the simpler default for the demo. (ADR-04.)
L2.3	3 NAT Gateways (one per AZ)	Cost vs blast-radius; Mittelstand-correct. Single-NAT designs are cheaper but break the AZ-isolation story we are otherwise telling.
L2.4	Cold DR via Velero + EBS snapshot	Hot-DR is structurally the wrong fit for LevelDB physics + 5-min cadence; see

#	Choice	Reasoning
		docs/operations/why-cold-dr.md. (ADR-04.)
L2.5	Master AZ rotation via aws eks update-nodegroup-config + Velero restore (not terraform apply)	Topology is constant; AZ choice is a runtime scaling decision. (ADR-01 modified, ADR-04.)
L2.6	3-tier flow: ALB → API → Envoy → StatefulSet, all in master AZ	Edge / business logic / mesh / state separation; standard senior shape.
L2.7	10 thematic ADRs (originally 50 private) — DR + LDB-layout topics consolidated into ADR-04	Submission covers structural depth, not surface coverage.

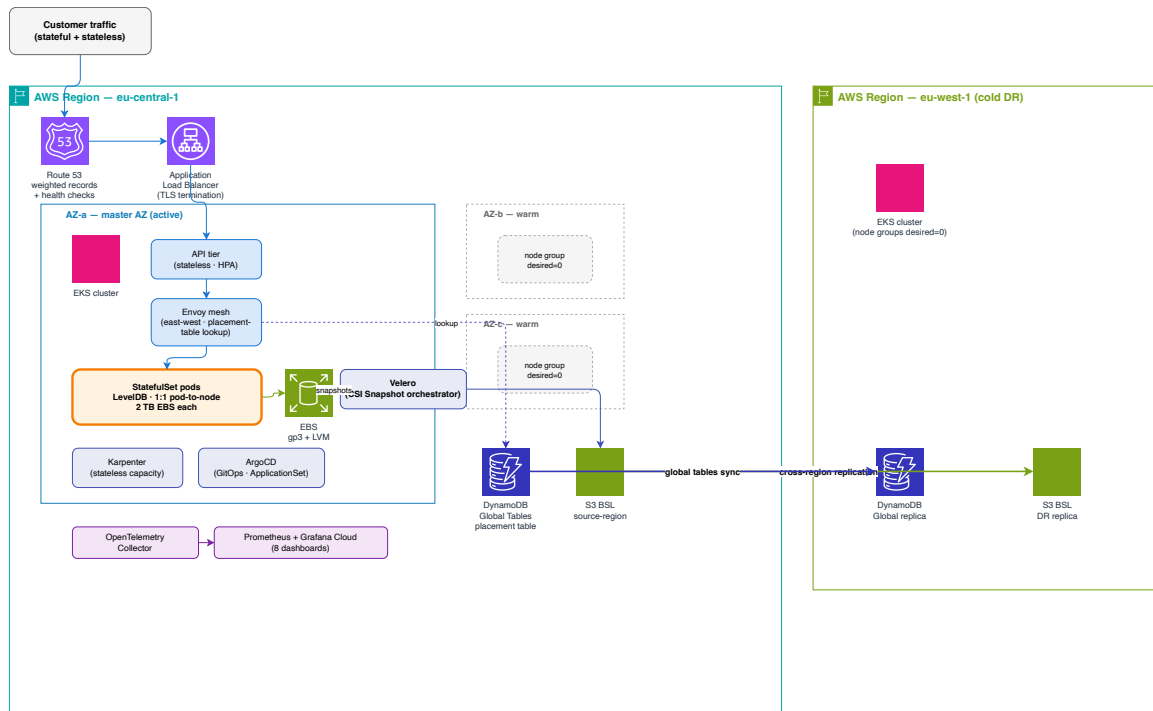
4. Customer configuration knobs (Layer 3 — operator decides)

The chart exposes the trade-off curve through a small, deliberate set of values. Full inventory in `helm/aegis-statefulset/values.yaml`.

Knob	Default	Range	Trade-off
<code>backup.cadence_minutes</code>	5	1 – 360	Lower = tighter RPO + higher EBS Snapshot S3 cost; upper bound is the spec's 6 h RPC
<code>ha.model</code>	<code>cold_dr</code>	<code>cold_dr</code>	Single mode in Wave 4. active passive (private ADR-014) retired in favor of cold DR (see ADR-04).
<code>routing.backend</code>	<code>dynamodb</code>	<code>dynamodb</code> , <code>customer_supplied</code>	DynamoDB Global Tables POC reference customer can substitute any backend that satisfies the 6-property contract per ADR-03.
<code>stateful.cells.count</code>	1	1 – N	POC default 1; production scale matches customer's

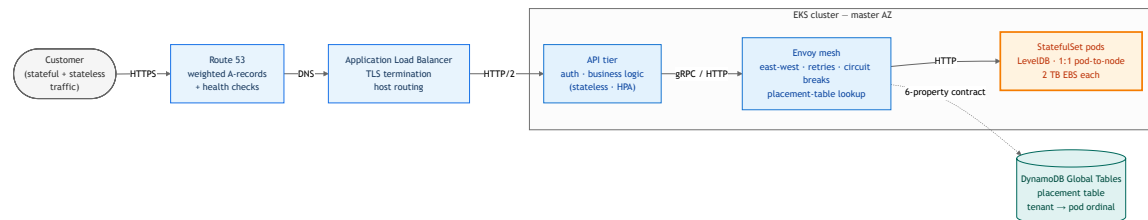
Knob	Default	Range	Trade-off
			existing partitioning per ADR-05 Strangler Fig migration.
storage.ebs_size_gb	2048	512 – 16384	Per-pod EBS size; LVM-managed and online-expandable.
master_az	eu-central-1a	any AZ in region	Master AZ for ALL workload Standby AZs desiredSize=rotation via aws eks update-nodegroup-config (ADR-01 modified, ADR-04).
dr_region	eu-west-1	any AWS region	Cross-region L target for EBS Snapshot copy via DLM (ADR-04).
observability.backend	grafana_cloud	grafana_cloud, amp_amg	OpenTelemetry (OTel)-instrumented; backend reversible per ADR-06.

5. Architecture diagram



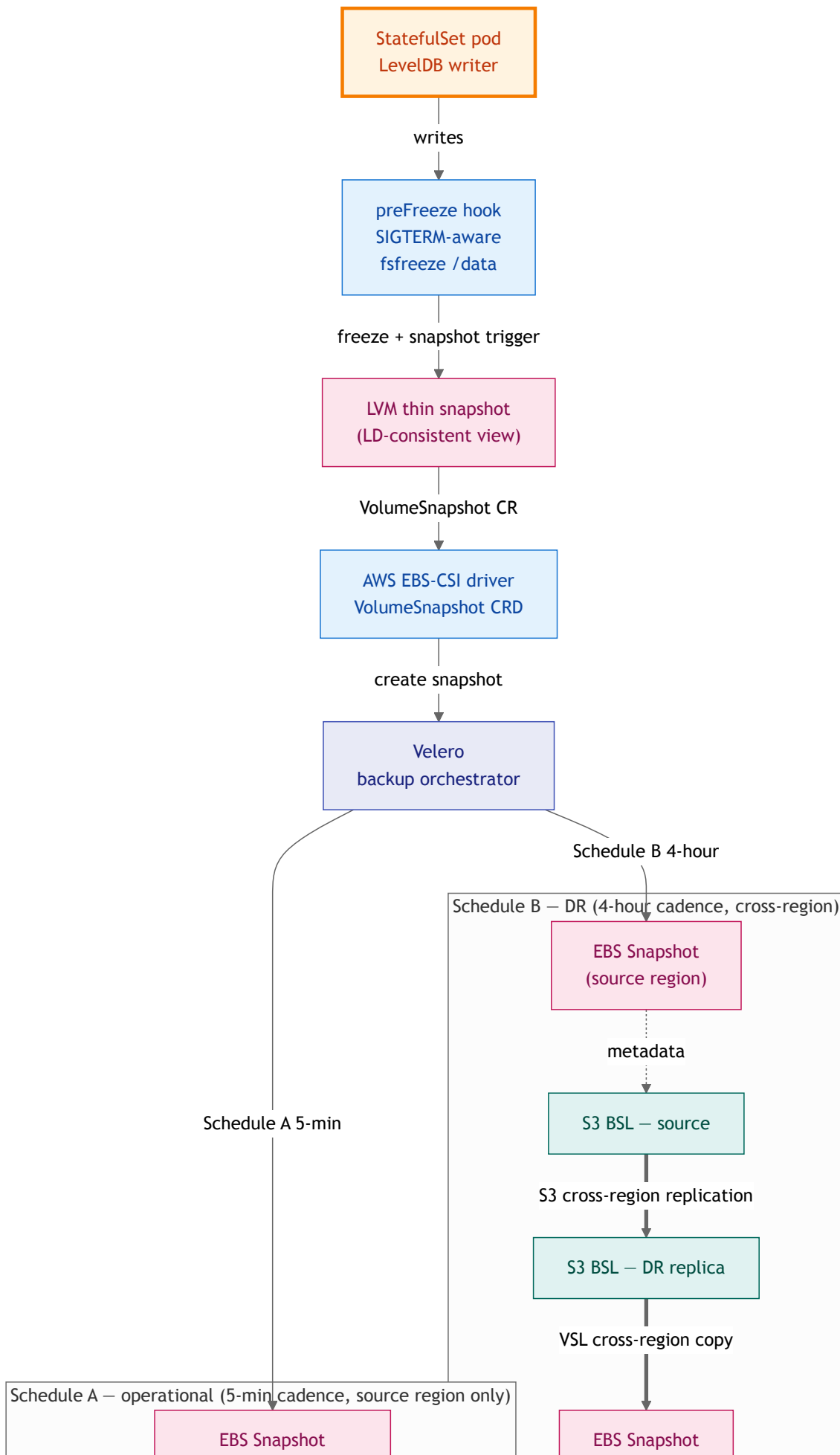
The diagram captures six structural choices on one canvas: (1) single master AZ for the stateful tier with AZ-b / AZ-c warm-standby on `desired=0`; (2) the three-tier ingress flow `ALB → API → Envoy → StatefulSet`, with the placement-table lookup pinned at the Envoy layer so storage backends stay swappable; (3) Velero as backup orchestrator on the CSI Snapshot path, with chunks landing in the source-region Backup Storage Location (BSL — Velero's S3-bucket abstraction) and AWS S3 cross-region replication keeping the DR-region BSL fresh; (4) ArgoCD deploying every workload from chart + values, no manual `kubectl apply`; (5) Karpenter handling stateless capacity while StatefulSet stays pinned on the master-AZ node group; (6) cold DR in eu-west-1 with node groups at `desired=0`, replicated BSL bucket, and DynamoDB global-tables replica waiting for the cutover runbook.

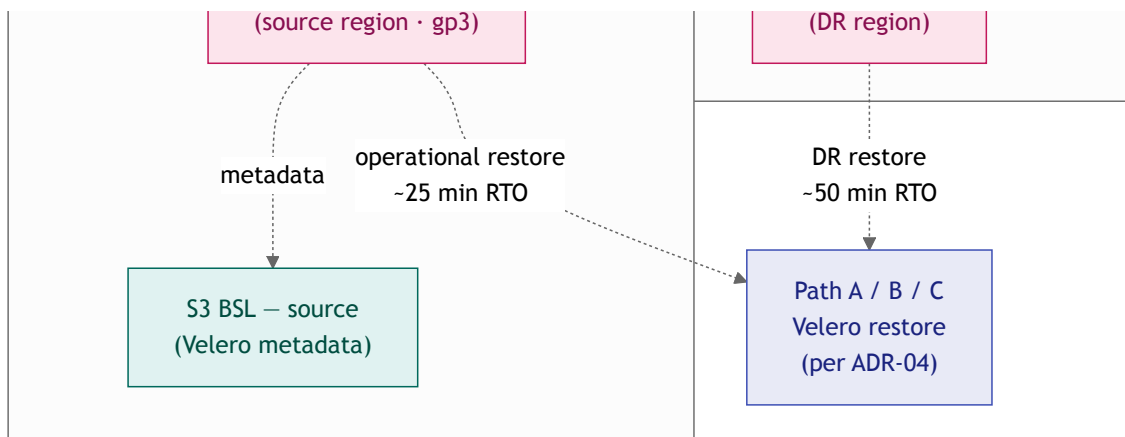
5a. Three-tier ingress flow (detail)



ALB does TLS termination + host-routing. API tier handles auth and business logic (stateless, autoscaled). Envoy carries the placement-table lookup against DynamoDB Global Tables and acts as the blue/green seam for the stateful tier. StatefulSet pods are pinned 1:1 to nodes in the master AZ. The 6-property contract (atomic CAS, strongly consistent reads, multi-AZ durable, low latency, CDC, audit log) is satisfied by DynamoDB in the POC; the customer can swap any backend that meets it.

5b. Backup data flow (detail)

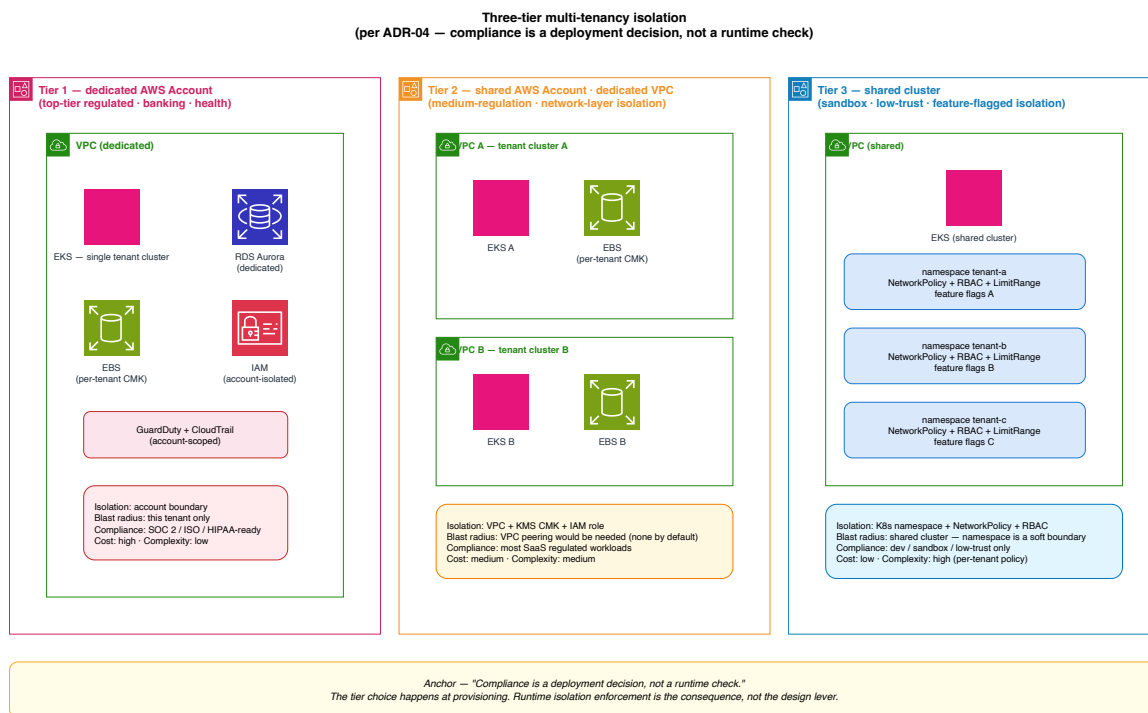




The backup pipeline starts at the Velero pre-hook that triggers the application's admin /quiesce endpoint (mock: sync && echo no-op; production: flush MemTable + WAL + drop file lock) to give LevelDB a stable view; the CSI driver creates a VolumeSnapshot CR per PVC (both data and wal PVCs per the multi-PVC pattern, atomically within a single Velero Backup CR); Velero orchestrates the underlying EBS Snapshot at the AWS storage plane. Two schedules land at different cadences: Schedule A (operational, 5-min, source-region only) and Schedule B (DR, 4-hour, source plus cross-region S3 replication plus VolumeSnapshotLocation (VSL) cross-region snapshot copy).

(LVM thin snapshot — the spec-literal pattern — is available as an opt-in path per docs/future/lvm-init-patch.md; modern K8s default delegates consistency to CSI VolumeSnapshot at the storage plane.)

5c. Three-tier multi-tenancy isolation (detail)



The architecture exposes three isolation tiers per ADR-04. Tier 1 (dedicated AWS account) is the top-tier regulated boundary — banking, health, anything subject to SOC 2 / ISO / HIPAA. Tier 2 (shared account, dedicated VPC) is the medium-regulation default — most SaaS workloads land here. Tier 3 (shared cluster, namespace + NetworkPolicy +

RBAC) is sandbox / low-trust only. The anchor sentence — *"compliance is a deployment decision, not a runtime check"* — captures the design lever: the tier choice happens at provisioning, runtime enforcement is the consequence.

3-tier flow detail: ALB does TLS termination + routing-by-host. API tier handles auth + business logic. Envoy provides east-west traffic shaping (retries, circuit breaks) and acts as a blue/green seam for the stateful tier. StatefulSet pods are pinned 1:1 to nodes in the master AZ.

6. Disaster recovery summary



Three failure shapes, three paths, each with named RPO / RTO.

Failure shape	Path	RPO	RTO	Mechanism
etcd corrupted, EBS intact	Path A	0	~30 min	EBS-tag truth → regenerate PV manifests with claimRef → Helm install at discovered count. (ADR-04, scripts/dr/recover-cluster.sh.)
Routing table lost, pods intact	Path B	0	~10 min	Pod inventory truth → re-derive override delta. (ADR-04, scripts/dr/warm-routing-table.sh.)
AZ failure (subnet, network, hardware)	AZ rotation	~5 min	~20 min	Standby AZ scale-up + Velero restore. (ADR-04, scripts/dr/az-rotation.sh.)
Source region lost	Path C / Region recovery	~5 min	~50 min	Cross-region cold-DR restore + Route 53 cutover. (ADR-04, scripts/dr/region-failure-recovery.sh.)
Backup lost too	Path C — last resort	bounded by cadence (max 6h per spec)	6–12h	Restic restore from cross-region S3. (ADR-04, scripts/dr/restore-from-s3.sh.)

The Three-Layer DR model (ADR-04) decouples these:

- **Layer 1 (Velero, cold)** — application namespaces.
- **Layer 2 (Terraform helm_release, warm)** — controllers (kube-system, monitoring, kyverno, External Secrets Operator (ESO), ArgoCD).

- **Layer 3 (Route 53, DNS)** — traffic.

Each layer recovers a different failure class without contention.

Live demonstration: Layer 1 (pod identity → PV re-attach) verified end-to-end on the staging cluster — see docs/evidence/ for the chaos-drill log (scripts/chaos/demo-stateful-pod-kill.sh, pod kill to Ready in ~65 s with both EBS volumes re-attached) and the CloudWatch corroboration (audit log events + per-volume IO metrics from the AWS side).

7. Cost summary

Component	Monthly (default)	Notes
EKS control plane	\$73	Single cluster
Stateful nodes (master AZ only)	\$500	r6id.2xlarge on-demand, 1:1 pod-to-node, POC <code>cells.count=1</code> ; scales linearly with cell count
Stateless nodes (api + envoy + system)	\$800	Karpenter mixed spot 70% / on-demand 30%
EBS gp3 storage (master AZ)	\$200	1× 2 TB primary at POC scale; warm-standby AZs charge \$0 (no PVCs until rotation)
3× NAT Gateways	\$99	Per-AZ blast-radius isolation; warm-standby rotation readiness (ADR-10 §6)
EBS Snapshot (5-min cadence, 30-day retention)	\$300	Velero schedule + DLM lifecycle
Cross-region snapshot copy (Glacier Instant Retrieval)	\$150	DLM cross-region per ADR-04
S3 (Velero metadata + cross-region replication)	\$50	Backup metadata only; data lives in EBS Snapshots
Observability (Grafana Cloud Pro)	\$500	OpenTelemetry-instrumented; Amazon Managed Prometheus / Amazon Managed Grafana (AMP/AMG) alternative ~\$700
Total (cold DR, 5-min cadence, POC <code>cells.count=1</code>)	~\$2,700	Mittelstand budget; scales linearly with cell count

Cost knobs:

- Loosen cadence 5-min → 1h → save ~\$200/mo, RPO worsens to ~30 min.
- Reduce to 1 NAT Gateway → save ~\$66/mo, lose warm-standby rotation readiness.
- Add cells per ADR-05 migration → ~\$700/mo per cell (node + EBS).
- **DR-posture cost dial** — see ADR-10 § 7. The "warm standby ~\$1,500/mo" alternative to cold DR breaks into a four-rung cost-per-capability ladder (Layer 0 cold / Layer 1 control-plane only / Layer 1+2 with minimal nodes / Layer 1+2+3 full warm), each rung with derivation, AWS pricing citation, and marginal \$/min saved.

The dial is the lever the customer moves if the RTO tolerance shifts — without rearchitecting.

All cost figures derive from formulas in docs/operations/cost-estimate-methodology.md — AWS pricing pages cited per line, methodology spelled out per component. Numbers shown as ~\$X/mo are approximations applied at POC scale; operator substitutes the customer's actual scale variables to derive their own number. Validate against AWS Cost Explorer before committing.

7a. Operational overhead — honest accounting

The architecture trades complexity for capability. This section names the operational cost so the trade is visible, per spec evaluation criterion #5 (*"awareness of performance, cost, complexity AND OPERATIONAL OVERHEAD"*).

What the SRE team carries

Surface	Cost
Three-Layer DR (Terraform infra / Helm-via-tf controllers / Velero application)	Three tools to operate, each with its own state model. Drift in one layer is invisible to the other two. Mitigated by docs/operations/scope-boundaries.md ownership contract + quarterly DR drill (ADR-04).
Six runbooks for human-side operations (Docker build, AWS bootstrap, OIDC, Grafana, submission email, INDEX)	Documentation discipline is a forever cost. Mitigated by runbook-per-flow shape — operators don't need to memorise; they read and execute.
50 private predecessor ADRs + 10 public mega-ADRs	Doc maintenance burden grows non-linearly. Mitigated by docs/adr/INDEX.md + the "private as history record, public as current architecture" split. The 10 public ADRs are the active surface; the 50 private are append-only history.
Pre-commit hook with 5 checks (anonymisation / Wave 1 stale / credential leak / kubeconform CRD-aware / hallucination defense two-tier severity)	Adds 5–10 sec to every commit. Mitigated by the hook being skippable for emergency commits via <code>--no-verify</code> (with the operator explicitly accepting the risk).
CI freshness check for the FSB diff patch (idempotent auto-detect)	Adds ~30 sec per PR for upstream-side patch validation. Mitigated by auto-detection of "downstream applied" state — customer who flips the patch doesn't have to manually disable.
Manual prod sync GitOps (per ADR-08)	Operator must approve every prod change; not auto-merged. Mitigated by — this is intentional, not overhead. The cost <i>is</i> the safety property.
Master AZ rotation discipline (per ADR-04 / ADR-15)	All three AZ node groups must stay patched + ready even if only one ever serves. Mitigated by Renovate auto-PRs; manual approval but low cognitive load.

Why we judge the overhead is carriable

A small platform team — call it 2–3 SREs sharing on-call — can carry this shape because:

1. **The three-layer separation is also the on-call separation.** A 3 a.m. page is most often "Layer 3 — application restart" (90 % of incidents, per usual SaaS distribution); operator goes to one runbook, one tool. Layer 1 incidents (terraform-level, e.g. region cutover) are quarterly events with a planned-event runbook. Layer 2 incidents (controller upgrade) follow the maintenance-window discipline.
2. **Documentation discipline compounds rather than burdens.** Each ADR the team writes today is one less re-derivation of context next quarter when a new engineer joins. The 50 private predecessors are the receipts that make the 10 public ADRs honest; pruning the 50 would reduce maintenance burden in the short term and cost auditability in the long.
3. **The FSB patch + CI freshness check is opt-in cost.** If the customer never flips the patch, the cost is one CI step running ~30 sec per PR. If they do flip it, the patch was already validated against the moment they applied it.
4. **The pre-commit hook + GitOps discipline replaces fire-fighting.** Five minutes of pre-commit friction prevents the half-day post-mortem when a brand-name leak or a stale Wave 1 string reaches production.

What we'd cut for a one-engineer platform team

If the customer's actual platform-team size is one engineer rather than 2–3, the architecture's overhead is too high. We'd cut, in priority order:

1. **Three-Layer DR → single-Layer Velero** — drop the terraform-managed helm releases for cluster controllers; use upstream Helm charts with ArgoCD instead. Trade-off: Layer 1 vs Layer 2 ownership contract weakens; easier for one engineer to operate, harder to audit.
2. **6 runbooks → 2** (chaos demo + submission email) — the rest become inline README sections. Trade-off: less ceremony, more risk of operator error.
3. **50 private + 10 public ADRs → 5 public ADRs** — collapse to the minimum that defends the architecture. Trade-off: future engineer joining the team pays the cost of re-deriving context.
4. **FSB patch + CI freshness check → drop entirely** — customer accepts "Velero CSI is the path; if you want Restic, fork the repo."
5. **Pre-commit hook → CI-only checks** — let the developer's first push fail fast in CI rather than locally. Trade-off: feedback loop is 60 sec slower per PR, but no local install burden.

These cuts shrink the architecture's surface by ~30 % without changing the core decisions (per-tenant pod / cold DR / placement table contract / defense-in-depth). The architecture is the same shape; the operational discipline scales to the team size carrying it.

Stage 3 conversation point

The right question for the panel is not "is this architecture good" — it's "can your team operate this architecture, and if not, which of the cuts above changes the answer?" The submission delivers the architecture; the panel decides which discipline level matches their actual organisational capacity.

8. Stage 3 questions consolidated

The submission deliberately defers these to a live conversation. Each maps to a knob, an ADR caveat, or an architecture variant.

1. **Service decomposition** — actual count and identity of microservices in the legacy stack?
2. **Routing key location** — tenant identifier in URL, subdomain, or JWT?
3. **Legacy session storage** — in-memory, file, Redis, or RDBMS?
4. **Placement algorithm** — deterministic by hash, or capacity-aware with a bin-packing component?
5. **Customer SLA shape** — single RPO across all tiers, or tiered (e.g., enterprise sub-2h cross-region RTO)?
6. **Multi-pod model** — confirm per-tenant cell vs hash-sharded?
7. **Pod density distribution** — many small tenants, or enterprise-dedicated profile dominant?
8. **Failover detection threshold** — 50% unhealthy / 2 min sustained acceptable?
9. **Failback policy** — Option A (standby becomes new primary permanently) vs Option B (rebalance back in maintenance window)?
10. **Maintenance window cadence** — what frequency is acceptable for operational rebalancing?
11. **LDB layout** — Pattern 1 (one DB shared by all tenants in pod) or Pattern 2 (one DB per tenant) — drives relocation tier matrix per ADR-04.
12. **Existing tenant→cluster mapping storage** — Postgres / Aurora / internal service / Redis?
13. **App code changeability** — can we add a small telemetry metric, or must the application container stay strictly unchanged?
14. **Migration window availability** — is there a planned window for Strangler-Fig cutover, or is it incremental?
15. **Customer's existing GitOps + DR tooling** — ArgoCD already in use? Velero already in use? — drives "what we own vs what we adopt".

Each maps to a knob in `values.yaml` or a caveat in an ADR. The submission gives the curve, not the answer.

9. Future work — "more time would add"

In rough priority order if the team wanted depth where the POC is shallow:

1. **WAL shipping** for sub-5-min RPO (out of POC scope; would change the active-passive design).
2. **Tier C relocation** — extraction code for live tenant relocation under Pattern 1 layout (~1 month of focused work per ADR-04).
3. **Per-tenant cost attribution dashboards** beyond the Athena queries in `scripts/finops/` — full Grafana panels with tenant-pivot tables.
4. **Multi-region active-active** — only if a customer SLA forces it; the default cold-DR + 50 min RTO is the right trade-off for Mittelstand.
5. **Service mesh upgrade** from Envoy as a sidecar to Cilium Service Mesh or Linkerd — east-west traffic shaping with kernel-level efficiency.
6. **Chaos automation** — schedule Phase 1 + Phase 2 drills as recurring GitHub Actions workflows with auto-roll-back; the scripts already exist, just need a scheduler.

7. **Per-tenant migration UX** — a small operator-facing CLI on top of `scripts/relocation/per-tenant-relocate.sh`.
8. **Backup verification automation** — periodic restore-and-checksum pipeline, beyond "did the snapshot land in S3".

Anything not in the POC is in this list because it is a depth choice, not a shape choice — the architecture supports each item without re-shaping.

10. For reviewers — entry points

You want to read about...	Start here
The whole picture	README.md and this file
Why each architectural decision	<code>_context/adr/</code> (50 private predecessors of the public 10)
DR-posture cost dial (FinOps tier ladder)	<code>docs/adr/ADR-10-finops.md</code> § 7
Why cold DR over active-passive multi-region	<code>docs/operations/why-cold-dr.md</code>
Disaster runbook	<code>docs/operations/region-failure-recovery.md</code>
Ownership boundaries	<code>docs/operations/scope-boundaries.md</code>
Per-tenant relocation tiers	<code>docs/operations/per-tenant-relocation.md</code>
Architectural assumptions	<code>docs/architecture-assumptions.md</code>
Mock app implementation	<code>app/main.go</code>
Chaos demo scripts	<code>scripts/chaos/</code>
Chaos demo + AWS observability evidence pack	<code>docs/evidence/</code> (chaos log + Grafana screenshots + CloudWatch JSON/PNG)
Helm chart	<code>helm/aegis-statefulset/</code>
Terraform	<code>infrastructure/terraform/</code>

Submission status: anonymised, public, intentionally portable. The architecture is reusable as a reference for stateful K8s workloads at Mittelstand-scale B2B SaaS, regardless of whether this particular conversation converts.

1. Customer's public pricing page (verified 2026-05-10). URL in the customer-specific grounding annex sent alongside this submission.↩
2. Customer's public information-security page (verified 2026-05-10). URL in the customer-specific grounding annex.↩
3. Customer's public information-security page (verified 2026-05-10). URL in the customer-specific grounding annex.↩

4. Customer's public information-security page (verified 2026-05-10). URL in the customer-specific grounding annex.↵